

# **Algebraic Topology: a beginner's course**

**This lecture:**

**AlgTop10: More on graphs  
and Euler's formula**

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Lecture 10: More on graphs and  
Euler's formula





Now have  $V, E, F$  vertices, edges, faces

$$3F = 2E \Leftrightarrow 6F = 4E$$

$$V - E + F = 2 \Leftrightarrow 6E - 6F = 6V - 12$$

$$2E = 6V - 12$$

Sum of all degrees is  $2E$

$$\text{So average degree} = \frac{2E}{V} = \frac{6V - 12}{V}$$

$$= 6 - \frac{12}{V} < 6$$

So at least one vertex is  
of degree or less!

vertices, edges, faces

$$\Leftrightarrow 6F = 4E$$
$$6F = 6V - 12$$

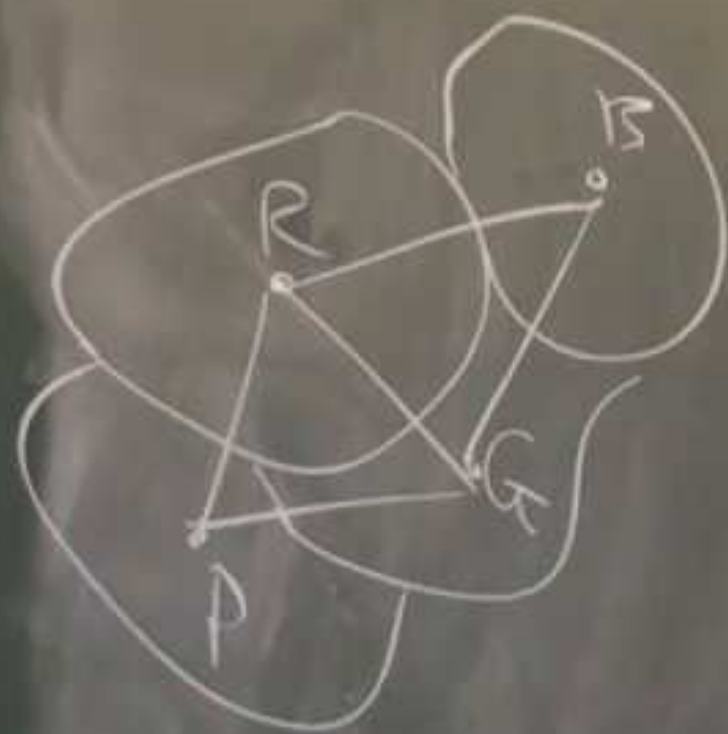
$$is \ 2E$$

$$\frac{2E}{V} = \frac{6V - 12}{V}$$

at least one vertex is  
of degree five or less

Thm. (6-colour thm) Every planar  
map can be coloured with six  
or fewer colours.

(Appel & Haken 1976 4 colours suffice)

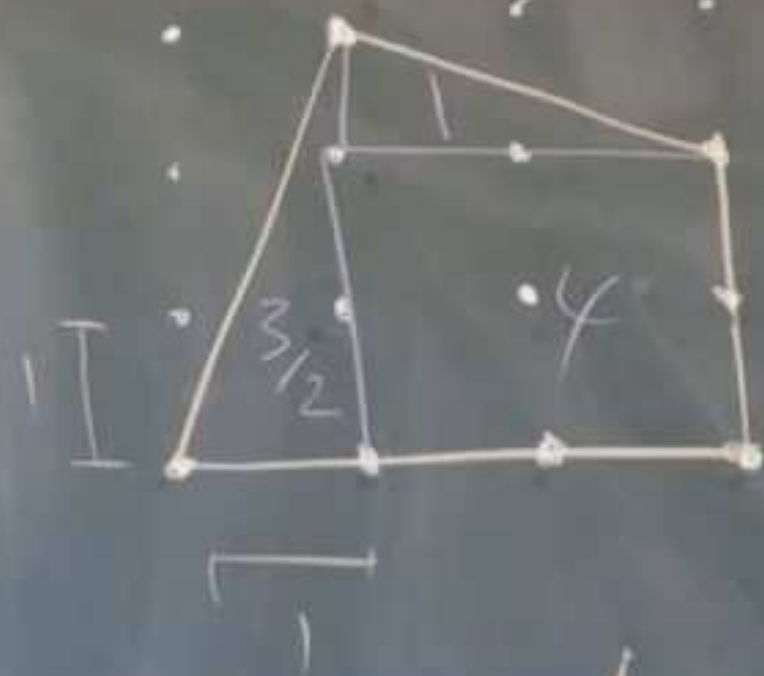


Graph version:  
planar graph can  
be vertex coloured  
with 6 colours so no 2  
adjacent ones have same colour.



# Pick's theorem

area of polygon } with integer vertices.



$$A = I + \frac{B}{2} - 1$$

$I$  = # interior points

$B$  = # boundary points

Ex.  $A = 4 + \frac{7}{2} - 1 = 6\frac{1}{2}$

$$\begin{pmatrix} 4 & 3 \\ 9 & 7 \end{pmatrix}$$

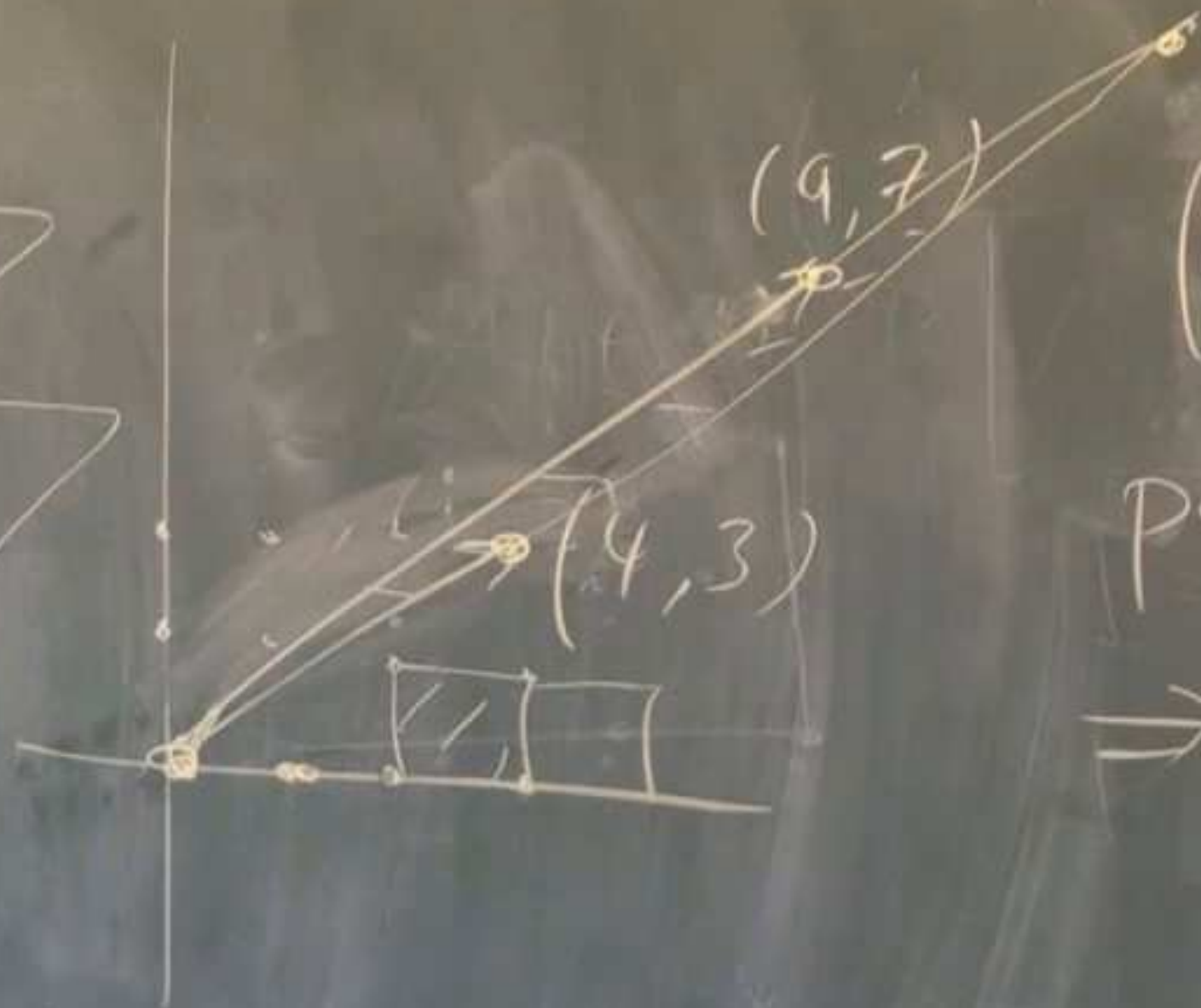
(c)





$$\begin{pmatrix} 4 & 3 \\ 9 & 7 \end{pmatrix}$$





$$\begin{pmatrix} 4 & 3 \\ 9 & 7 \end{pmatrix}$$

parallelogm has 4 int. points

$$\Rightarrow \text{area} = 1$$



Euler's formula:

$$\chi = V - E + F$$

$$= (I + B) - \left( \frac{2I}{2} + \frac{B}{2} \right) + \left( \frac{2I}{2} + \frac{B}{2} \right)$$

$$= I + B - I - \frac{B}{2} + I + \frac{B}{2}$$

So  $\chi = I + B$  and  $A = I$

**AlgTop11: Rational curvature, winding and turning**

Proof  
convex  
polyg

Subd  
 $F =$

# **Algebraic Topology: a beginner's course**

**next in the series:**

**AlgTop11: Rational curvature,  
winding and turning**